



What is a Mole?

One of the 7 base units in the SI system is the **mole**, which refers to the amount of substance, be it atoms, molecules or particles.

A mole (abbreviated as mol) is defined as the number equal to the number of carbon atom in exactly 12 grams of pure ¹²C.

$$1 \text{ mole} = 6.022 \times 10^{23}$$

The Avogadro's Number

6.022×10^{23} is called the **Avogadro's number**, named after the Italian mathematician and scientist **Lorenzo Amadeo Avogadro**. This means that for every mole of a specific element, say, Hydrogen, there should be 6.022×10^{23} atoms of the same element.

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Molar Mass

Molar mass is the mass of one mole of a substance. Naturally, the atomic mass of a substance in grams per mole (g/mol) or in atomic mass unit (amu) or in unified atomic mass unit (u) is also its molar mass (g). For example, Carbon has an atomic mass of 12 g/mol or 12 amu or 12 u. Therefore, its molar mass should be 12 g.

$$\begin{aligned} 1 \text{ u} &= 1 \text{ amu} \\ &= 1 \text{ dalton} \\ &= 1 \text{ g/mol} \\ &= 1.66 \times 10^{-27} \text{ kg} \end{aligned}$$

Using these relationships, an amount of a substance in moles or in atomic mass units can be converted to its equivalent mass in grams (or kilograms) and vice versa.

Molecular Mass

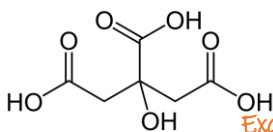
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Molecular mass is the sum of atomic masses in the molecule. To get the molecular mass, one should multiply the atomic mass of an atom to the number of similar atoms present in the molecule. For instance, if you want to get the molecular mass of water (H₂O), then just get the sum of the atomic mass of two hydrogen atoms (1 amu + 1 amu) and the atomic mass of one oxygen atom (16 amu), giving you a molecular mass of 18 amu. Naturally, the molecular mass should be equal to the molar mass of the compound.



Molecular weight of common substances in u

Substance	Molecular Mass (u)
Hydrogen gas (H ₂)	2
Methane (CH ₄)	16
Water (H ₂ O)	18
Oxygen Gas (O ₂)	32
Methyl Alcohol (CH ₃ OH)	32
Butane (C ₄ H ₁₀)	58



Chemical Formulas

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Chemical formulas are used to designate the atoms present in different molecules and compounds are these formulas show the ratio in which they combine.

- Molecular Formula** – this type of chemical formula shows the exact number of atoms from each element that is present in the molecule or the compound. For example, H₂O is the molecular formula of water. It means that to form water, one atom of oxygen must be chemically combined with two atoms of hydrogen.
- Empirical Formula** – this type of chemical formula shows a simplified ratio of the atoms from each element that is present in the molecule or compound. Unlike the molecular formula, this will not show the exact number of atoms present, thus it is the simplest chemical formula. For example, HO is the empirical formula of hydrogen peroxide with molecular formula H₂O₂. There are instances, however, when the empirical formula and molecular formula of one molecule or compound are one and the same. For example, water has a molecular formula and an empirical formula of H₂O.

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Percent Composition

The percent composition by mass gives the amount in terms of percentage of an element in a compound.

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$$\% \text{ composition} = \frac{\text{mass of an element in the compound}}{\text{total mass of the compound}} \times 100\%$$



Gases

Gases exist in nature under certain conditions, pressure and temperature.

Most of these gases behave in such a way that is more predictable compared to solids and liquids. Different theories and laws have been formulated to quantify this behavior of gases. Though it is sometimes used interchangeably with “vapor”, gases and vapors are actually different. Gases exist in its gaseous form at room temperature. Vapors, on the other hand, exist in liquid form at room temperature and will only be transformed to gas by increasing the temperature of the substance.

Characteristics of gases

- Gases follow the volume and shape of their container
- Gases have lower densities compared to solids and liquids
- Gases have molecules that are far apart; hence it moves randomly and freely, unlike other phases of matter. These molecules are always in motion.
- Gases can be more easily compressed compared to solids and liquids.
- Different gases will mix evenly if they are found on one container
- Some gases are toxic and poisonous (e.g. hydrogen sulfide) while others are essential to sustain life (e.g. oxygen gas)
- Most gases are colorless although some are visible (e.g. nitrogen dioxide is visible in polluted air with its brownish color)
- Gases exert pressure on surfaces that are in contact with them.

Gas Pressure

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The standard unit of pressure is **Pascal** (Pa) or Newton per square meter (N/m²). Atmospheric pressure, first noticed by the Italian scientist Evangelista Torricelli, is the pressure exerted by the earth's atmosphere. Since air is denser near the surface of the earth, pressure should be higher near the surface. Atmospheric pressure depends on temperature and the altitude or location. Standard atmospheric pressure at 1 atm is the pressure that supports a 760-mm column of mercury at 0°C at sea level. This can be tested using a barometer, a common device used to measure atmospheric pressure. Another device used to measure pressure other than the atmospheric pressure is the manometer.